The Spiral of Hypotheses

The founding field: nine working wagers for living infrastructure in formation

Foundation paper · founding synthesis / current orientation · candidate v0.4.1 · 11 October 2026
Scope: independent inquiry into the Association as a first bounded institutional field · Accountable steward: Lars A. Engberg

Publication activates nothing. AI supported articulation, comparison and structural synthesis; situated authority and responsibility remain human.

What this paper is

This paper distils a larger public working register of sixty field hypotheses and four meta-hypotheses. The public *Protocol Habitat* describes how foundations, protocols, applications and tools relate. This paper offers the founding figure beneath that architecture; it does not activate a protocol or demonstrate that the wagers hold elsewhere.

A founding field, not a finished model

Spiralweb Stewardship Association is more than the legal holder of a ledger, and less than a centre entitled to govern a distributed field. It is the present institutional ground for a wider public habitat: Green Papers hold shared inquiry and public memory; Planetary Guardians develops governance and protocol architecture; AnchorPoints hold relational continuity; Fields hold operational status and consequence; Methods connect situated practice and evidence; and the PG Ledger may preserve memory, provenance and correction where that is useful.

The association itself is the first place where this architecture must become answerable. Its governance, membership, publication practice, use of AI, treatment of paid and unpaid work, receipt of support, documentation burden and capacity to show its own yellow and red conditions are not administrative matters outside the inquiry. They are part of the inquiry.

In that bounded sense, the association is establishing a **living laboratory for the supporting conditions** under which regenerative communities, public knowledge and responsible support may meet. It tests its own capacity and conduct first. People and places connected through Spiralweb are not experimental subjects, and a relationship does not become a field test

merely because the association describes it. Any situated application begins through legitimate local mandate and retains the right to translate, correct, refuse, pause or leave the shared form.

This is therefore a founding field: not a territory, a pilot site or a claim that the whole architecture is active, but an accountable institutional practice in formation. The question is whether the association can help life-serving knowledge, relationships and resources meet without making itself sovereign over them.

Growing like a food forest

A food forest offers more than an attractive metaphor for this formation. It is a layered practice of succession: soil, water, roots, canopy, understory, human work and more-than-human life develop at different tempos while affecting one another. Its form is neither a monoculture nor a finished blueprint installed all at once. What matters is whether relationships among the layers become more capable of holding life over time.

The parallel is deliberately incomplete, but it helps make the architecture visible:

IN THE FOOD-FOREST FIGURE	IN THE SPIRALWEB ARCHITECTURE
Ground and soil	Ethical, constitutional and epistemic foundations
Roots and living connections	Relationships, AnchorPoints and locally held continuity
Different layers and functions	Places, practices, knowledge, governance, tools and support without one sovereign layer
Water, nutrients and human tending	Money, attention, care, labour and institutional capacity moving through honest membranes
Observation across seasons	PG Ledger memory, evidence, uncertainty and later consequence where a record is useful
Pruning, decomposition and succession	Correction, pause, retirement, endings and revised form
Seeds and independent growth	Practices and shared disciplines that can travel through translation without making the centre heavier

The comparison does not require every place to become a food forest or adopt one land-management school. Food forests, farms, wetlands, school gardens, municipalities and human–AI inquiry are different contexts. The common discipline is succession rather than

replication: begin from actual conditions, add no more than can be held, observe consequence, and let later form remain open.

Three movements of establishment

The founding field can be understood through three recurring movements. They are not a maturity ladder imposed on local places, a timetable, or gates that publication moves anyone through. Several movements may coexist, repeat or stop.

Grounding — prepare the conditions. Name the wagers, establish constitutional and legal ground, preserve provenance, clarify authority, distinguish document status from field status, and form relationships strong enough to carry honest correction.

Rooting — become answerable to consequence. Where a legitimate relationship invites it, make one bounded shared movement; keep local articulation intact; ask what can be observed without extraction; test whether the record, support and coordination are useful; and let steward viability limit expansion.

Propagation — let warranted learning travel. When a function survives situated correction, offer it as a revisable paper, protocol or tool whose claim boundary remains visible. Translation is not replication. Successful propagation means more distributed capacity, not necessarily more organisation at the centre.

The public *Protocol Habitat* describes the corresponding spiral in operational terms: generic material may meet an AnchorPoint, become a situated application, encounter consequence, leave a record where appropriate, and return as correction. One application may be enough to disprove a general claim. It is almost never enough to validate a universal one.

How this relates to the First and Second Spirals

In this paper, the spiral names the recurring movement through which wager, practice, consequence and correction return to one another. The First and Second Spirals on the Papers site are publication cycles within that wider movement: the constitutional cycle and the applied inquiries. They are related, but they are not the same classification, and neither is a maturity ladder.

Why offer wagers rather than principles?

Spiralweb's papers, reports, protocols and operational formats approach a recurring set of questions from different directions. What can a body, a relationship or an institution actually carry? How can support move without becoming control? What kinds of knowledge become visible when observation begins in a place rather than at a distance? What should remain humanly accountable when AI contributes to public knowledge work? And what institutional forms can connect situated practices without erasing their differences?

The texts do not resolve into a doctrine. They form a figure: a small number of linked orientations that can guide inquiry only if they remain open to correction. We call them **working wagers** because something is placed at stake—time, attention, relationships, institutional design—and because the claims may lose. A wager here is not a prediction made for gain. It is an orientation stated clearly enough that experience, evidence or criticism can require us to change it.

The nine wagers are not all hypotheses in the same scientific sense. Some are empirical propositions. Others are design hypotheses, normative commitments, definitions or interpretive syntheses. Their correction conditions therefore differ. A claim may need to be narrowed, divided, reclassified or abandoned rather than simply verified or falsified. Correction may come from situated practice, stewards, scholarship, public evidence, legal review, critical friendship or better reasoning. No field relationship or reporting duty is created by publishing this paper.

The revision conditions do not all operate at the same timescale. Some may be opened by one consequential event; others can only be judged comparatively across repeated cycles, different settings and, in some cases, years. A single event may require correction without settling the wager, while a durable pattern may require revision even where no one event is decisive. These conditions guide inquiry; they do not automate verdicts.

The three opening wagers form one movement: 0-a concerns who can now build public knowledge infrastructure; 0-b asks how situated knowing may connect into democratic correction; and 0-c names the living reality to which knowledge, authority and institutional action must remain answerable.

The figure in one movement

> **Ethics concerns what we can carry, not only what we believe.**
> → **Capacity is the ability to remain coherent while changing.**
> → **Capacity makes critical friendship possible:** close enough to understand, independent enough to question, committed enough to want the work more truthful.
> → **Critical friendship depends on restored distinctions:** support is not control, fluency is not understanding, guardianship is not ownership, and participation is neither capture nor opposition.

This movement is the proposed spine of the nine wagers. If its root claims do not hold, much of the figure must be reconsidered. Naming the dependency makes the paper more vulnerable to correction, not less.

The living is not invoked here as a sovereign exception or as a substitute for evidence. It names the bodies, relationships, places and more-than-human conditions to which every abstraction must remain answerable. Rule of Life remains within Rule of Law; care does not suspend rights, scrutiny or public accountability.

Wager 0-a · The threshold has fallen here: situated practice can become public knowledge infrastructure without surrendering human mandate

The threshold that has fallen is not the threshold for producing more text. It is the threshold between lived, situated inquiry and coherent public infrastructure. Here, one named human steward, using AI under a human mandate, has held together memory, retrieval, comparison, structural synthesis, editorial formation, technical publication and revision across a body of work that would ordinarily require a research group, an editorial function, technical capacity and an institutional home. AI helped carry the exogenous labour of comparison and structural sight; the human remained responsible for purpose, situated judgement, relationship and the Last Impulse.

Spiralweb is therefore more than evidence that an output can be produced. It is a situated existence proof that institutional legibility need not begin with institutional scale. If this capacity transfers, small and place-bound practices may be able to develop their own public

memory, methods and correctable architectures, and connect through shared grammar without transferring substantive authority to a centre. Universities, public institutions, consultancies and large organisations would retain important capacities, but their practical advantage in determining what can become coherent, durable and publicly legible would become less exclusive. That possibility—not the volume of this corpus—is the radical claim.

The same threshold falls for care and manipulation. The capacities that allow a situated steward to build a correctable knowledge habitat also allow an individual or small group to produce the appearance of institutional authority—volume, coherence, citation and polished public form—without plural scrutiny, durable procedures or meaningful accountability. As more people work this way, public knowledge may become more distributed and more difficult to trust at the same time. Coherence can no longer be treated as evidence of truth, legitimacy or human judgement.

The wager therefore does not make institutions unnecessary. It changes the question of what they are for. When institutional form becomes widely producible, the indispensable functions are provenance, contestability, rights protection, continuity, correction and named human responsibility. These functions may be held through renewed institutions, shared protocols and polycentric public bodies, but they cannot be supplied by fluency itself. The lowered infrastructural threshold makes the human Last Impulse more consequential, not less.

This demonstration does not yet show that the practice transfers, distributes authority, remains viable, or works equitably across bodies, languages and places. **Revision would be required if** the apparent gain proved to be speed or volume rather than durable infrastructure; if hidden labour, inaccessible tools, displaced responsibility or exceptional personal burden explained the result; if other stewards could participate only through a progressively heavier centre; or if the proliferation of AI-made institutional form weakened public judgement, contestability and accountability more than it distributed situated capacity.

Opens into full-register H0-a, H33, H33b and H50, and into the coordination and accountability problems taken up by Wagers 0-b and 5.

Wager 0-b · Shared grammar can distribute democratic sensing and constitutional capacity without conferring mandate

If the infrastructural threshold described in Wager 0-a continues to fall, citizen science may become more than participation in research. It may become a democratic practice through which people can observe, interpret and contest the consequences of public decisions from

where those consequences are lived. A person may begin with a bounded place, encounter or institutional output; distinguish observation from interpretation, hypothesis and decision; use AI to read legislation, budgets, plans, evidence and prior commitments; and formulate a precise, corrigible public claim. The observation is knowledge at its own scale. It is not thereby a decision, a vote or a mandate to govern.

The 13×13 materials offer one bounded grammar for this movement. Their purpose is not to partition reality or make every person use the same categories. They allow questions about body, place, power, relation, time, evidence and correction to be approached from different positions without requiring one position to become the sovereign interpreter of the others. Questions may travel while answers, ownership and authority remain situated. A public decision can therefore be opened into inspectable propositions, mandates and expected effects, then encountered through many concrete pixels of consequence. The pixels do not add up to the whole field, but they may reveal patterns, contradictions and harms that an aggregated institutional account cannot see.

An election or lawful appointment may confer a legitimate and time-bounded mandate to decide. It does not confer a monopoly on knowing what decisions do. Citizen science can add a continuing correction layer between elections: people become not only voters, consultees or sources of data, but observers, knowledge-holders, correctors, co-interpreters, initiators and participants in shared priority-setting. One precise observation may open a serious inquiry without becoming sovereign over the outcome. Many observations may disclose a pattern without volume alone becoming proof. Institutions remain necessary to test claims, protect rights, hear disagreement, give reasons, decide within lawful mandates and remain answerable for what follows.

As more people work this way, democratic knowledge may become more polycentric. Local circles may compare observations across time and place; shared ledgers may preserve provenance, uncertainty, disagreement and response; AI may assist translation and comparison; and elected assemblies, courts, public institutions, professional bodies and civic networks may receive a more precise account of how common decisions meet lived reality. The radical possibility is not continuous polling or the replacement of representative democracy. It is distributed constitutional capacity: more people and places becoming able to understand, test and help correct the order that acts upon them, without each becoming a miniature sovereign.

The same capacity can deform democracy. AI can manufacture persuasive accusations, apparent constituencies and thousands of institutionally polished pixels at negligible cost. Coordinated campaigns can make repetition appear to be independent observation. Shared grammars can homogenise experience, platforms can turn participation into surveillance, and model or protocol owners can become hidden constitutional centres. Citizen science can

become unpaid public administration, extract local knowledge or expose people and places that never consented to visibility. Local circles can reproduce inherited domination, while the language of polycentricity makes exclusion harder to see. Connection is not consent; legibility is not legitimacy; distributed capacity is not distributed mandate.

Revision would be required if this practice did not make consequential public error easier to detect or correct than existing democratic and administrative channels; if shared grammar repeatedly homogenised places or transferred interpretive authority to a model, platform or centre; if participation became surveillance, unpaid administrative labour or synthetic mobilisation; if volume displaced precision, rights or evidence; if unmandated networks acquired substantive authority over affected people; or if distributed constitutional capacity weakened the ability of legitimate institutions to make, explain and repair common decisions.

Opens into full-register H0-b, H0-d, H20–H24b, H34, H38, H38b, H39, H50 and H54, and into the situated-knowledge and human-accountability problems taken up by Wagers 3 and 5.

Wager 0-c · The living field can be a primary unit of attention and consequence, while responsibility remains named

Public decisions become real in living fields. A law, budget, plan, grant, technology or organisational strategy may be authorised in one place, administered in another and experienced across many others. Its consequences appear in bodies, relationships, workloads, water, soil, other species, local economies, institutional trust and the future capacity of a place to sense and respond. The living field names these relations among place, people, more-than-human life, knowledge, money and governance. It does not replace projects, organisations or legal entities. It makes visible the reality their boundaries leave outside.

Situated observation and citizen science give the field ways to answer back. A decision's propositions and expected effects may be encountered through bounded, revisitable pixels: a plot of land, a service encounter, a school route, a digital process, a workplace relation or another site of consequence. Repeated observations can be compared, questioned and connected without pretending that the field is the sum of its records. A blank may be legitimate. Some realities may be witnessed but never collected. The unrecorded, those who do not consent to visibility and the not-yet-visible remain part of the field.

Attention to the field also changes how value and accountability are read. An efficiency gain is not complete if work has merely moved to citizens, families or another institution. Ecological improvement cannot compensate for a steward being consumed by the work. A valuable dataset does not excuse the loss of local authority over knowledge. Financial, ecological,

human and governance effects should therefore remain separately visible. Accountability must be able to follow money, body, relation and consequence across organisational boundaries, while preserving the distinction between what was observed, how it was interpreted, who decided and what later occurred.

The living field is not a person, a legislature or a legal accountability-holder. It has no single voice. To say that a place has agency is to say that it responds, resists, enables and alters what becomes possible; it is not to say that it speaks with one intention or issues a human-readable mandate. People may testify to those responses from situated positions, but every account remains an interpretation with provenance, uncertainty and a standing right of contestation. No individual, organisation, model or local circle may therefore claim to speak for the field in full. Attention to the field does not suspend individual rights, rule of law, professional evidence, due process, democratic representation or the protection of minorities and dissenters. A field may be a primary unit of attention, consequence and value-reading; responsibility remains with named humans and legitimate institutions that possess the mandate and capacity to act.

The field-language carries its own danger. It can mystify causation, turn political choices into apparently organic necessities and allow those with power to speak as though "the field" had authorised them. More-than-human life can be ventriloquised. Local tradition can be romanticised while domination remains hidden. A demand for holistic legibility can become total surveillance, and an AI-generated representation can be mistaken for the living reality it only partially describes. If everyone is said to be responsible to the field while no decision-maker remains answerable, the language of field accountability conceals a responsibility gap.

Revision would be required if attention to the living field repeatedly obscured rather than clarified who caused, authorised or could repair a consequence; if project-, organisation- or rights-based units tracked relevant effects as well as or better than the field framing; if holistic observation required extractive data collection or made people and places excessively transparent; if appeals to local or more-than-human authority weakened universal rights, lawful duties or the standing of dissenters; if the burden of making institutions answerable fell primarily on those already carrying the consequences; or if field representations became more influential than the realities and people capable of correcting them.

Opens into full-register H0-c, H28–H32, H48–H49 and H55–H56, and into the non-compensatory, situated-knowledge, support and viability problems taken up by Wagers 2, 3, 4 and 6.

Wager 1 · Ethics concerns capacity, and capacity concerns regulation through change

People may know what ought to be done and still be unable to do it. Sustained stress narrows attention, shortens time horizons and degrades judgement. Ethics therefore concerns not only declared values but what bodies, relationships and institutions can carry. Regulation is used here to mean the capacity to remain sufficiently coherent while changing—not obedience, discipline or the suppression of conflict.

The wager operates at three levels. Bodies need conditions in which attention and response remain possible. Relationships need enough trust and boundary to hold disagreement. Institutions need ways to pause, learn and alter course without hiding damage. The practical question is not simply whether an action sounds good, but whether it gives to life without requiring what the people and relations carrying it cannot sustainably provide.

This is partly a conceptual framing and partly an empirical claim. It should not be inflated into a universal biological law, and it cannot by itself decide what justice requires.

Revision would be required if durable ethical judgement reliably remained intact under sustained dysregulation at body, relational and institutional levels; if the language of regulation concealed coercive normalisation; or if capacity proved too vague to distinguish legitimate limits from avoidance of responsibility.

Opens into full-register H1–H8 and H36.

Wager 2 · Different readings should remain separate and non-compensatory

A place cannot be adequately read through one number. Spiralweb separates at least three readings: Land/Life, Steward Viability, and Governance/Commons. They illuminate different relations and should not be averaged into a composite score. Ecological strength cannot cancel an exhausted steward; sound governance cannot erase harm to land; human viability cannot make institutional opacity disappear.

To **read, not score** is not to reject measurement. It is to resist false commensurability. Colours, thresholds and categories are documented human judgements, not properties that reality carries by itself. Each judgement should retain its evidence, uncertainty, author and date. An honest gap is more informative than a fabricated green.

Non-compensation also has limits. Public decisions sometimes require prioritisation, aggregation and comparison. The wager is that these should be made visibly and politically rather than concealed inside one apparently objective score.

Revision would be required if a composite measure repeatedly guided sound decisions without hiding burdens or trade-offs; if separated readings produced paralysis without greater honesty; or if users could not understand the distinctions well enough to act on them.

Opens into full-register H8, H8b and H16–H19.

Wager 3 · Situated knowledge is genuine, fallible and not exhaustive

Observation from within a place is knowledge, not raw material waiting for validation from above. Bodies, relationships and repeated presence can register changes that remote systems miss. Situated knowledge should therefore enter public and institutional reasoning in its own right.

But situated does not mean complete or beyond criticism. Every position reveals and obscures. Expert assessment, legal evidence, remote sensing, scientific research and the testimony of others may correct local accounts; local observation may in turn expose the abstractions and blind spots of distant expertise. The task is not to crown one sovereign way of knowing but to build accountable interfaces among different, internally diverse forms of evidence.

Knowledge from a field should ordinarily be offered, not extracted. Any shared copy should have a clear purpose, provenance and accountable holder. Lawful reporting, safeguarding duties and fiduciary minimums can justify required information; they should be specified rather than hidden beneath a claim that all sharing is voluntary. Some domains may rightly remain witnessed without becoming collected data.

Revision would be required if mandatory upward data flows repeatedly proved necessary, proportionate and non-extractive in practice; if situated accounts proved no better than distant ones at detecting relevant change; or if withholding information repeatedly prevented lawful protection, learning or accountability.

Opens into full-register H20–H24b and H58.

Wager 4 · Support should not buy substantive local control

Money is never just money. It can carry a legitimate purpose, legal conditions, safeguarding requirements, financial-accountability duties and expectations about agreed work. Those conditions are not automatically capture. The critical boundary lies between accountability for support and purchase of substantive control over local governance, knowledge, land, story or priorities.

An honest membrane separates money, purpose and decision. It makes conditions explicit; identifies who can decide what; records conflicts of interest; protects the possibility of refusal; and asks whether pace and reporting burden fit the field's actual capacity. Guardianship, access and authorship should not quietly become ownership.

The wager does not promise that membranes remove power. Funders retain choices, legal entities retain duties, and scarcity shapes what can be refused. The claim is narrower: making these relations visible and limiting the authority carried by support may reduce capture.

Revision would be required if support that openly purchased governance or veto reliably strengthened local capacity without silencing knowledge or shifting burdens; if the proposed separation obscured rather than clarified actual power; or if lawful, proportionate conditions could not be distinguished in practice from substantive control.

Opens into full-register H26–H30 and H51.

Wager 5 · Fluency is not understanding, and responsibility remains human

AI can produce fluent language, detect patterns and assist synthesis. Fluency is not evidence of first-person experience, moral standing or situated understanding. Questions about machine interiority can remain philosophically open without making practical responsibility obscure.

For consequential AI-assisted work, a named human should hold the mandate, examine the sources, make the judgement and remain answerable for correction and consequence. That person must be more than a ceremonial sign-off. Human accountability is meaningful only where the holder has time, competence, access and authority to intervene.

This paper follows that discipline. AI assistance is acknowledged; authorship and responsibility are not transferred to the tool. The living realities described here are not required to align

with a machine's categories. The categories must remain revisable in contact with bodies, relationships, places, evidence and law.

Revision would be required if processes without a named human holder proved equally or more accountable over time; if naming a person consistently created responsibility theatre rather than real scrutiny; or if distributed accountability could be specified in a form that protected judgement and remedy better than individual stewardship.

Opens into full-register H33, H33b and H50.

Wager 6 · Viability vetoes expansion; roles rotate and pace follows living capacity

A system that depends on permanent heroes is already transferring its costs into bodies. When steward viability reads red, expansion pauses. This veto applies to new ambition, added reporting demands and growth; it does not erase lawful duties, safeguarding responsibilities, emergency action or obligations already incurred. Roles should rotate where possible; exposed positions should have relief; and work should be allowed to pause, contract or end. Calendar regularity is not proof of health. Money, ambition and reporting cadence should adapt to the metabolism of the people and relations carrying the work.

Viability is not comfort. Some roles involve enduring duties; emergencies compress time; public institutions must sometimes act despite fatigue. The design question is whether necessary strain remains named, bounded, shared and repairable—or becomes the hidden operating system—and whether fresh ambition is stopped before it consumes the capacity needed to meet existing responsibility.

Ending can also be intelligent. A project or relation that has completed its work may need decomposition rather than indefinite preservation. Continuity should be earned by continuing value and capacity, not by attachment to organisational survival.

Revision would be required if systems repeatedly remained just and viable while keeping the same people at the exposed edge; if forced institutional tempo reliably regenerated living systems; or if rotation and pause displaced responsibility without distributing competence.

Opens into full-register H12–H15, H25, H41 and H57.

What holds the wagers together

Critical friendship is the proposed method. It is a relation close enough to understand, independent enough to question, and committed enough to seek greater truthfulness. It is neither opposition, capture, consultancy nor surveillance. It appears across the architecture: between steward and field, human and AI, citizen and institution, and Spiralweb and the economies it seeks to question. Calling it the overarching method is an interpretation of the corpus, not a quotation from it, and remains open to revision.

Distinction-restoration is the tactic. Boundaries are not necessarily violence. Support is not control. Guardianship is not ownership. Publication is not activation. Fluency is not understanding. Restoring such distinctions makes it possible to stay close without being captured and remain independent without becoming an enemy.

Correction has a spiral form. A place or institution returns to a living question changed by what happened. Shared language can help different places learn from one another, but cannot carry their judgement for them. The spiral is an interpretive image, not a claim that all learning follows one geometry.

Open questions

These are not hidden inside the wagers as if already solved.

- **Conflict and adjudication between rightful fields.** Withdrawal from a voluntary relationship does not resolve conflicts over rights, resources or shared consequences. Legitimate common institutions remain necessary.
- **The missing macroeconomic categories.** What would an ecologically honest public economy need to count, and which existing categories obscure living costs? This remains a research programme, not a finished model.
- **Legitimate binding and the uninvited critic.** How can a Rule-of-Life test deepen ecological accountability while remaining within Rule of Law, universal rights, due process and institutions whose scrutiny does not depend on invitation?
- **Reflexivity winners and losers.** AI may lower some thresholds while creating new exclusions around access, language, education, safety, time and regulatory capacity. Who gains the ability to question binding rules, and who remains spoken for?
- **The ontology of the AI relation.** Whether human–AI practice needs a positive account beyond practical safeguards remains open. The nine wagers do not depend on resolving machine interiority.

How to read and respond

Read the nine as a connected public hypothesis, not as a checklist. Ask first where the wording overreaches, where different claim types have been confused, and which people or forms of evidence have been excluded. Then ask what would make the claims more precise, more just and more usable.

The full public working register preserves the source map, the sixty field hypotheses, the four-part meta-layer and their individual correction lines. This public paper is a door into that depth and a description of the association's own founding inquiry. It does not activate a field, appoint a steward, impose a protocol, create reporting duties or claim a relationship with any place or institution.

The invitation is therefore specific: **What do these wagers help us see? What do they hide? What evidence, experience or argument should make us revise them?**

Suggested citation: Engberg, Lars A. (2026). *The Spiral of Hypotheses: The founding field — nine working wagers for living infrastructure in formation.* Foundation paper, candidate v0.4.1. Spiralweb Papers. https://papers.spiralweb.earth/papers/the-spiral-of-hypotheses. CC BY 4.0.